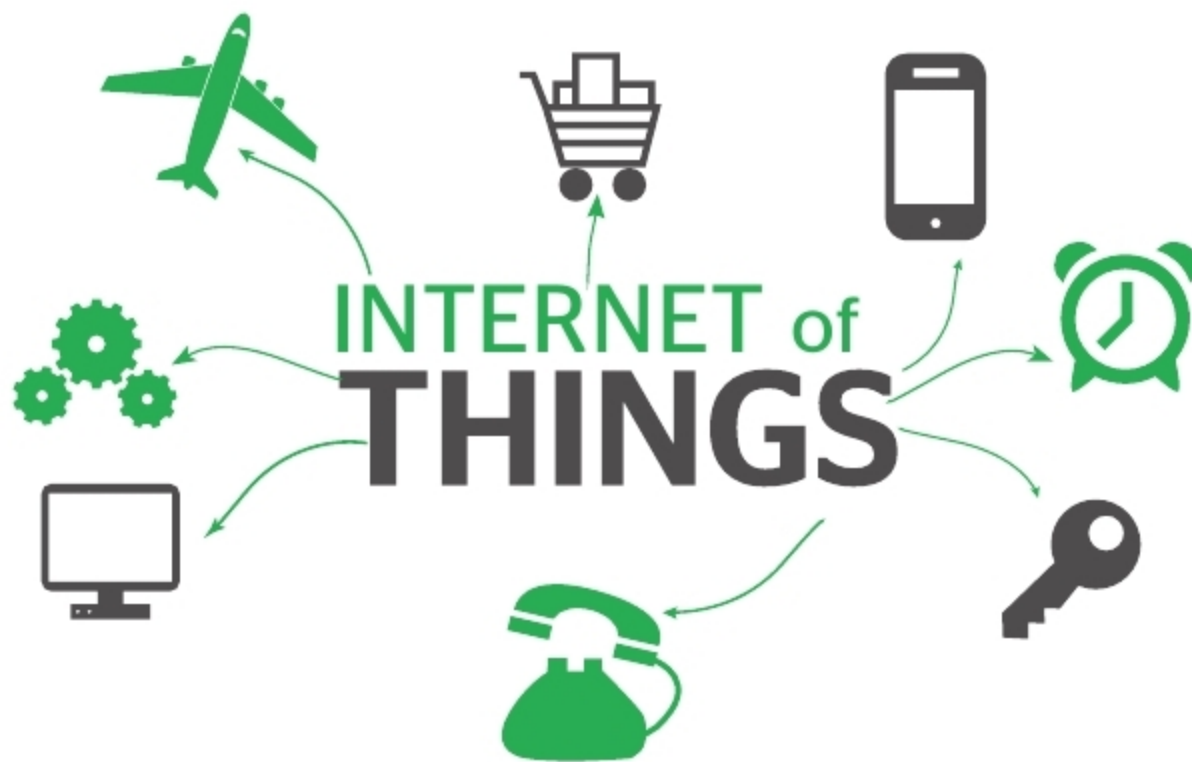


The Protocols That Help Things to Communicate Over the Internet

The Internet of Things is a system of connected, interrelated objects. These objects transmit data to servers for processing and, in turn, receive messages from the servers. These messages are sent and received using different protocols. This article discusses some of the protocols related to the IoT.



- **Mobile app:** Using a mobile application, the user can view and access the data processed in the cloud server.

This is the basic idea of the IoT and its architecture, along with the components. We now move on to the basic ideas behind different IoT protocols.

IoT protocols

As mentioned earlier, connected things are used to collect information, convey/send information back, or do both, using the Internet. This is the fundamental basis of the IoT. To convey/send information, we need a protocol, which is a set of procedures that is used to transmit the data between electronic devices.

Essentially, we have two types of IoT protocols — the IoT network protocols and the IoT data protocols. This article discusses the IoT data protocols.

MQTT

The Messaging Queuing Telemetry Transmit (MQTT) protocol was primarily designed for low bandwidth networks, but is very popular today

The Internet of Things (IoT) is beginning to pervade more and more aspects of our lives. Everyone, everywhere is using the Internet of Things. Using the Internet, *connected* things are used to collect information, convey/send information back, or do both. IoT is an architecture that is a combination of available technologies. It helps to make our daily lives more pleasant and convenient.

IoT architecture

Basically, IoT architecture has four components. In this article, we will explore each component to understand the architecture better.

- **Sensors:** These are present everywhere. They help to collect data from any location and then share it to the IoT gateway. As an example, sensors sense the temperature at different locations, which helps to gauge the weather conditions. And this information is shared or passed to the IoT gateway. This is a basic

example of how the IoT operates.

- **IoT gateway:** Once the information is collected from the sensors, it is passed on to the gateway. The gateway is a mediator between sensor nodes and the World Wide Web. So basically, it processes the data that is collected from sensor nodes and then transmits this to the Internet infrastructure.
- **Cloud server:** Once data is transmitted through the gateway, it is stored and processed in the cloud server.

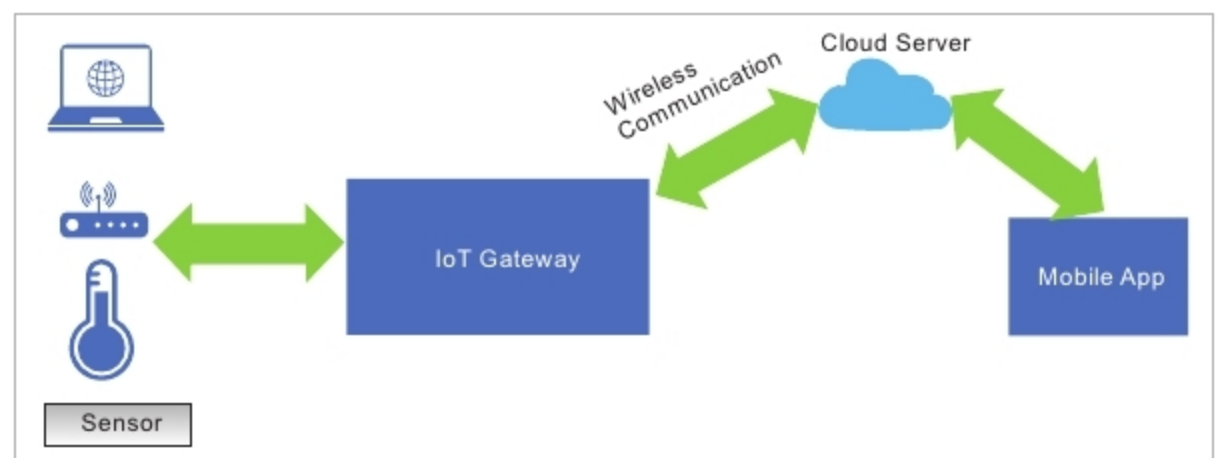


Figure 1: IoT architecture